MAT357
Foundations of Real Analysis

Class Lecture Notes

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Contents

Preface **ii**

I Introduction and Preliminaries **1**

1 Review, Metric, and Normed Vector Spaces 1

2 Metric Space Topology 5

II Metric Spaces of Functions **15**

3 Basics 15

4 Arzela-Ascoli Theorem 19

5 Convolutions and Weierstrass Approximation 21

6 Equicontinuity and Stone-Weierstrass 26

7 Two Metric Space Theorems 30

8 Fourier Series 36

III Measure Theory **46**

9 Basics 46

10 Limits of measurable functions 59

11 Simple functions 62

Preface

These notes were created during class lectures. As such, they may be incomplete or lacking in some detail at parts, and may contain confusing typos due to time-sensitivity. Additionally, these notes may not be comprehensive. Most statements in this document which are not Theorems, Problems, Lemmas, Corollaries, or similar, are likely paraphrased to a certain degree. Please do not treat any material in this document as the exact words of the original lecturer.

If you are viewing this document in Obsidian, you may notice that the links in the pdf document do not work. This is intentional behaviour, as I currently do not have or know of a decent solution which allows them to behave well with the setup in Obsidian. However, below certain pages, there may be links to other documents - these are usually context-relevant links between notes of different areas of study. I created these links to point out potential similarities, or in case one area of study is borrowing a concept, definition, or theorem from another area of study, and you wish to see the full, original definition/derivation/proof or whatever it may be.

I Introduction and Preliminaries

1 Review, Metric, and Normed Vector Spaces

Lec 1 - Jan 6 (Week 1)

we start w some review, particularly of $\mathbb{R}$. recall the least upper bound property: for any $S \subseteq \mathbb{R}$, M is an upper bd for S if for all $x \in S$ we have $x \leq M$.

fact: any nonempty $S \subseteq \mathbb{R}$ bdd above has a least upper bd.

Theorem 1.1: Archimedean Property

If $a < b \in \mathbb{R}$ are distinct, then $\exists\, q \in \mathbb{Q}$ such that $a < q < b$.

Proof. Source: Primary Source Material

sps wlog $0 < a < b$. let $M \in \mathbb{N}$ such that:

$$M > \frac{1}{b-a} \implies M(b-a) > 1$$

let $N \in \mathbb{N}$ be the largest s.t. $N \leq Ma$. then $q = \frac{N+1}{M}$ satisfies $a < q < b$. indeed:

$$N+1 > Ma \implies a < \frac{N+1}{M} \qquad Mb > Ma + 1 \geq N+1 \implies b > \frac{N+1}{M}$$

■

genuinely hes just reviewing 157. why. at least hes moved on to metric spaces at [checks watch] 9:47. its now 10:00 and hes defining a nvs.

Lec 2 - Jan 8 (Week 1)

ℓ^p spaces. we know these. note $p \in [1, \infty]$.

$$\ell^p = \left\{ (a_n) : \sum_n |a_n|^p \right\} < \infty, p < \infty \qquad \ell^\infty = \{(a_n) : \sup_n |a_n| < \infty\}$$

we know these norms. check ℓ^p is a vector space:

$$|a_n + b_n|^p \;\leq\; (|a_n| + |b_n|)^p \;\leq\; (2\max(|a_n|, b_n))^p \;\leq\; 2^p(|a_n|^p + |b_n|^p)$$
$$\implies \sum |a_n + b_n|^p < \infty$$

now we claim p-norm is a norm. most important is triangle inequality.

Lemma 1.2: Young's Inequality

if $x, y \geq 0$, then:

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q}$$

for conjugate exponents $p, q \in (1, \infty)$.

Proof.

exercise; consequence of the fact that $s \mapsto \ell^s$ is cvx. apparently.

Theorem 1.4: Holder's Inequality

sps $p, q \in [1, \infty]$ s.t. $1/p + 1/q = 1$ (conjugate exponents). let $a \in \ell^p$, $b \in \ell^q$. then:

$$\sum |a_n b_n| \leq \|a\|_p \|b\|_q$$

Proof. Source: Primary Source Material

case $p = 1, q = \infty$. then:

$$\sum |a_n b_n| \leq \sum |a_n| \sup_n |b_n| = (\sup_n |b_n|) \sum |a_n| = \|a\|_1 \|b\|_\infty$$

case $p, q \in (1, \infty)$: wlog, assume $\|a\|_p = \|b\|_q = 1$. apply young's:

$$|a_n b_n| = |a_n||b_n| \leq \frac{|a_n|^p}{p} + \frac{|b_n|^q}{q}$$

$$\sum |a_n b_n| \leq p^{-1} \sum |a_n|^p + q^{-1} \sum |b_n|^q = \frac{\|a\|_p^p}{p} + \frac{\|b\|_q^q}{q} = p^{-1} + q^{-1} = 1$$

■

Theorem 1.5: Minkowski's Inequality

let $p \in [1, \infty]$, $a, b \in \ell^p$. then $\|a + b\|_p \leq \|a\|_p + \|b\|_p$.

Corollary 1.6

ℓ^p is a nvs.

Proof. Source: Primary Source Material

$p = 1, \infty$ easy, so sps $p \in (1, \infty)$. then:

$$\sum |a_n + b_n|^p = \sum |a_n + b_n||a_n + b_n|^{p-1} \leq \sum |a_n||a_n + b_n|^{p-1} + \sum |b_n||a_n + b_n|^{p-1}$$

note:

$$p^{-1} + q^{-1} = 1 \implies q^{-1} = 1 - p^{-1} = \frac{p-1}{p}$$

apply holder's to first sum:

$$\sum |a_n||a_n + b_n|^{p-1} \leq \|a\|_p \left(\sum |a_n + b_n|^{(p-1)q} \right)^{1/q} = \|a\|_p \left(\sum |a_n + b_n|^p \right)^{1-p^{-1}}$$

applying to both sums gives:

$$\sum |a_n + b_n|^p \leq (\|a\|_p + \|b\|_p) \left(\sum |a_n + b_n|^p\right)^{1-p^{-1}} = \left(\|a\|_p + \|b\|_p\right) \|a+b\|_p^{p-1}$$

$$\|a+b\|_p^p \leq (\|a\|_p + \|b\|_p)\|a+b\|_p^{p-1} \implies \|a+b\|_p \leq \|a\|_p + \|b\|_p$$

■

Exercise 1.7 Source: Primary Source Material

prove that if $p < q$ then $\ell^p \subsetneq \ell^q$. hint: consider $\sum 1/n^s$ for some good s.

Example 1.8 Source: Primary Source Material

$C[0,1]$, the nvs of cts $f : [0,1] \longrightarrow \mathbb{R}$. p-norm for $p \in [1, \infty]$ entirely analogous.

Exercise 1.9: HW 1.4 Source: Primary Source Material

repeat holder, minkowski pfs to show:

$$\|fg\|_1 \leq \|f\|_p \|g\|_q \qquad \|f+g\|_p \leq \|f\|_p + \|g\|_p$$

"sequences and convergence" are we just doing topology again. well now we're doing continuity, specifically seq continuity. so yes, we're just doing topology again.

hint (idea?) for hw3, basically versions of 1-x but seq $\to$ more curve.

Lec 3 - Jan 13 (Week 2)

some equivs btwn continuity, bdries, i forgot what else. evidently im not rly paying attn oh equiv of metrics/norms. note that if $p < q$, then $\ell^p \subsetneq \ell^q$ so $\|\cdot\|_p$ and $\|\cdot\|_q$ not equiv.

2 Metric Space Topology

Lec 4 - Jan 15 (Week 2)

tdy was a snow day so going off posted notes but i still think nothings happened.

Example 2.1 Source: Primary Source Material

$C([0,1])$ is complete under sup norm.

let f_n be cauchy. for all $x \in [0,1]$, we have:

$$|f_n(x) - f_m(x)| \leq \sup_x |f_n(x) - f_m(x)| = \|f_n - f_m\|_\infty$$

thus, $(f_n(x))$ is cauchy for fixed x, so $f_n(x) \to f(x)$ as $n \to \infty$ for fixed x. want to show that $f \in C([0,1])$ and $\|f - f_n\|_\infty \to \infty$ as $n \to \infty$.

let $\varepsilon > 0$. then there exists $N > 0$ s.t. for all $n, m \geq N$, $\|f_n - f_m\|_\infty < \varepsilon$. since f_n is cauchy, for all x, we have for $n \geq N$:

$$|f(x) - f_n(x)| = \lim_{m\to\infty} |f_m(x) - f_n(x)| \leq \lim_{m\to\infty} \left(\sup_x |f_m(x) - f_n(x)| \right) < \varepsilon$$

thus for all $n \geq N$, $\sup_x |f(x) - f_n(x)| < \varepsilon$, so f_n cvgs uniformly to f.

now we show f cts. let $\varepsilon > 0$ and $x \in [0,1]$. then there exists N such that $\sup_y |f(y) - f_N(y)| < \varepsilon$. since f_N cts, $\exists \delta > 0$ s.t. $\forall y \in [0,1]$ w $|x - y| < \delta$, we have $|f_N(x) - f_N(y)| < \varepsilon$. thus:

$$|f(x) - f(y)| \;\leq\; |f(x) - f_N(x)| + |f_N(x) - f_N(y)| + |f_N(y) - f(y)| \;\leq\; 3\varepsilon$$

this concludes the proof.

lol ok thats it.

Lec 5 - Jan 20 (Week 3)

still more cpt stuff ...

Definition 2.2

we say $f : M \longrightarrow N$ is **uniformly cts** if $\forall \varepsilon > 0$, $\exists \delta > 0$ s.t. $\forall x, y \in M$, w/

$$d_M(x, y) < \delta \implies d_N(f(x), f(y)) < \varepsilon$$

notice the distinction btwn this and regular continuity.

Theorem 2.3

if $f : M \longrightarrow N$ cts and M cpt, then f uni. cts.

Proof. Source: Primary Source Material

pick a seq $(x_n, y_n) \in M^2$ s.t.

$$d_M(x_n, y_n) \to 0$$

since M^2 cpt, $\exists (x_{n_k}, y_{n_k}) \to (x, y)$. then:

$$d(x, y) \leq d(x, x_{n_k}) + d(x_{n_k}, y_{n_k}) + d(y_{n_k}, y) \implies d(x, y) = 0$$

so $x = y$. therefore:

$$d(f(x_{n_k}), f(y_{n_k})) \leq d(f(x_{n_k}), f(x)) + d(f(y), f(y_{n_k})) \implies d(f(x_{n_k}), f(y_{n_k})) = 0$$

since f cts. ■

oh hey connectedness

Lec 6 - Jan 22 (Week 3)

more connectedness ig

Theorem 2.4

$\mathbb{R}$ is conn

Proof. Source: Primary Source Material

let $U \subseteq \mathbb{R}$ be non-empty clopen. wts $U = \mathbb{R}$.

let $p \in U$ and defn:

$$X := \{x \in U : (p, x) \subseteq U\}$$

X non-empty since U open, $p \in U$. claim: $\sup X = \infty$.

sps otw, let $s := \sup X < \infty$. we show $s \in X$.

by defn, $\exists x_n \to s$. then $\forall n, (p, x_n) \subseteq U$, so:

$$(p, s) = \bigcup_n (p, x_n) \subseteq U$$

since $x_n \in U$ and U closed, $s \in U$ so $s \in X$. since U open, $s + \varepsilon \in X$ for all $\varepsilon > 0$.

thus, we must have $(p, \infty) \subseteq U$. by a symmetric argument, $(-\infty, p) \subseteq U$. we have $(-\infty, p) \cup \{p\} \cup (p, \infty) \subseteq U$, so $U = \mathbb{R}$. ■

"for culture".

one place conn is used: pdes, known as *method of continuity*.

generally, want to find soln to eq:

$$Lu := \sum a_{ij} \frac{\partial}{\partial x_i} \frac{\partial}{\partial x_j} u = f$$

given $f : \mathbb{R}^n \longrightarrow \mathbb{R}$, find u s.t. $Lu = f$. for(?) $a = (a_{ij})$ is elliptic, $\lambda^{-1} \leq ? \leq \lambda$ for some $\lambda > 0$. when $a_{ij} = \delta_{ij}$, ie $a = \mathrm{id}$, this gives poissons eq'n.

for $t \in [0,1]$, define:

$$L_t = t \sum_i \frac{\partial^2}{\partial x_i^2} + (1-t) \sum_{i,j} a_{ij} \frac{p}{\partial x_i} \frac{\partial}{\partial x_j}$$

$$I = \{t \in [0,1] : L_t u = f \text{ has soln}\}$$

wts $I = [0,1]$. let ?? $\in I$, try to show ??? clopen.

anyway.

Definition 2.5

let $S \subseteq M$. $p \in M$ is a **cluster pt** of S if $\forall r > 0$, $|B_p(r) \cap S| = \infty$.

Theorem 2.6

tfae

1. $\exists (x_n) \in S$ of distinct pts cvg to p
2. $\forall r > 0$, $|B_p(r) \cap S| = \infty$
3. $\forall r > 0$, $|B_p(r) \cap S| \geq 2$
4. $\forall r > 0$, $B_p(r) \cap S$ contains a pt $\neq p$

Proof. Source: Primary Source Material

clearly 1 $\implies$ 2 $\implies$ 3 $\implies$ 4. we show 4 $\implies$ 1.

let $x_1 \in B_p(1) \cap S$ s.t. $x_1 \neq p$. then $r_1 < d(x_1, p)$. let $x_2 \in B_p(r_1 \cap S)$ s.t. $x_2 \neq p$. note $x_2 \neq x_1$ since $r_1 < d(x_1, p)$. take $r_2 < \min(d(x_2, p), 1/2)$, induct. ■

Definition 2.7

a metric space is **perfect** if every pt is a cluster pt.

Theorem 2.8

complete non-empty perf metric sp is unctbl. (crll: $\mathbb{R}$ is unctbl)

Proof. Source: Primary Source Material

sps $M = \{x_1, x_2, \ldots\}$ ctbl. we construct a seq of closed sets of the form:

$$Y_i := \overline{B_{y_i}}(r_i)$$

s.t. $\forall n$:

1. $x_n \notin Y_n$
2. $Y_{n+1} \subseteq Y_n$
3. $r_n \leq 1/n$

assuming the construction, $(y_n)_n$ is cauchy. sps $y_n \to y$. by 2, since we have that $y_m \in Y_n$ $\forall m \geq n$, then $y \in Y_n$ $\forall n$. but $x_n \notin Y_n$ so $y \neq x_n$ $\forall n$, contradiction.

we now construct such a seq of sets. let $y_1 \in B(x_1, 1)$ s.t. $y_1 \neq x_1$. next, define $r_1 < \min(d(y, x_1), 1)$. then $Y_1 = \overline{B_{y_1}}(r_1)$ satisfies $x_1 \notin Y_1$.

for any k, choose $y_k \in B(y_{k-1}, r_{k-1})$ s.t. $y_k \neq x_k$. choose $r_k > 0$ s.t.:

$$r_k < d(y_k, x_k) \qquad r_k \leq 1/k \qquad B(y_k, r_k) \subseteq B(y_{k-1}, r_{k-1})$$

then $x_k \notin \overline{B}(y_k, r_k)$. ■

Example 2.9: Cantor Set

see below

as a fun ex of a perf space sth sth containing no interval, we construct the std middle thirds cantor set.

oh he drew some pictures

$$A_n := [0,1] \setminus \left(\bigcup_{j=0}^{n-1} \left(\frac{1+3j}{3^n}, \frac{2+3j}{3^n} \right) \right) \qquad C := \bigcap_n A_n$$

properties of C:

1. closed + cpt
2. "totally disconn"
3. non-empty
4. perf but contains no interval
5. unctbl

"totally disconn" means $\forall r > 0, p \in C$, $\exists$ set U clopen in C s.t. $U \subseteq B(p,r) \cap C$.

n.b.: "proper" defn is that only singletons are conn.

pf that C non-empty: consider endpts.

pf that C unctbl: closed bdd subset of $[0,1]$, so complete + perf thus unctbl.

pf that C contains no interval: sps $(\alpha, \beta) \subseteq C$. then if $1/3^n < |\beta - \alpha|$, C_n is intervals of len $1/3^n$, so $(\alpha, \beta) \subsetneq C_n$. since $C \subseteq C_n$, $(\alpha, \beta) \subsetneq C$.

pf that C totally disconn: let $r > 0, p \in C$. choose n s.t. $1/3^n < r$. then $p \in C_n$ implies $\exists$ interval I of len $1/3^n$ in C_n s.t. $p \in I$. wts $U = I \cap C$ satisfies:

- $U \subseteq B(p,r) \cap C$
- U clopen in C

since $U = I \cap C$, I closed interval, U is thus closed in C. write $I = [a,b]$ and choose $\varepsilon < 1/3^n$. then:

$$I \cap C \;=\; [a,b] \cap C \;=\; (a-\varepsilon, b+\varepsilon) \cap C \;=\; U$$

so U open in C.

pf that C perf: let $p \in C$ and $r > 0$. wts $|B(p,r) \cap C| \geq 2$. if n s.t. $1/3^n < r$, then $p \in I$ is a closed interval of len $1/3^n$ in C_n. [take endpts]

Lec 7 - Jan 27 (Week 4)

defn: cvring cpt: topo defn of cpt. thm: (seq) cpt iff cvring cpt iff complete + totally bdd. note he phrased the 2nd one as a set $\subseteq M$ being cpt iff *closed* + totally bdd, with M complete, rather than just complete + totally bdd.

Definition 2.10

given metric sp M, a space M' is a **completion** of M if:

i) $\exists \iota : M \longrightarrow M'$ isometry onto $i(M)$

ii) M' complete

iii) $\iota(M)$ dense

note: he uses $\widehat{M}$ but i dont feel like doing that so.

Theorem 2.11

every metric sp has a completion

Proof. Source: Primary Source Material

sketch of "real" proof. define the set of cauchy seqs in M as:

$$\mathcal{C} \ := \ \{(x_n) : x_n \in M, (x_n) \text{ is cauchy}\}$$

defn:

$$a \sim b \iff \lim d(a_n, b_n) = 0 \qquad M' := \mathcal{C}/\sim \qquad D([a],[b]) \ := \ \lim d(a_n, b_n)$$

we check:

i) lim exists and is well-defn'd

pf: skipped, exercise

ii) is a metric

pf: triangle [basically] gets inherited, nothing to do

iii) M' complete

pf: next time

iv) inclusion is an isometry

pf: inclusion given by constant seqs, isometry follows

v) img of inclusion is dense

pf: next time

■

Lec 8 - Jan 29 (Week 4)

last 2 pfs from last time

pf that M' complete: let $p_k \in M'$ be cauchy. for each k, choose a representative $p_k \in [(p_n^{(k)})_n]$ s.t. $d(p_n^{(k)}, p_m^{(k)}) \leq 1/k \ \forall n, m$. we can do this by modifying finitely many terms at the start of each seq. (?)

claim: $a_n = (p_n^{(n)})$ is cauchy and $D(p_k, [(a_n)_n]) \to 0$. let $\varepsilon > 0$. then $\exists N$ s.t. $\forall n, m \geq N$, $D(p_n, p_m) < \varepsilon$. lets also assume $1/N < \varepsilon$. then for $n, m \geq N$, we have that $d(a_n, a_m) = d(p_n^{(n)}, p_m^{(m)})$. for any $\ell \in \mathbb{N}$, we have:

$$\begin{aligned}
d(p_n^{(n)}, p_m^{(m)}) &\leq d(p_n^{(n)}, p_\ell^{(n)}) + d(p_\ell^{(n)}, p_\ell^{(m)}) + d(p_\ell^{(m)}, p_m^{(m)}) \\
&\leq 1/n + d(p_\ell^{(n)}, p_\ell^{(m)}) + 1/m \\
&\leq 1/N + 1/N + d(p_\ell^{(n)}, p_\ell^{(m)}) \\
&\leq 2\varepsilon + d(p_\ell^{(n)}, p_\ell^{(m)}) \\
&\leq 2\varepsilon + D(p_n, p_m) \text{ as } \ell \to \infty \\
&\leq 3\varepsilon
\end{aligned}$$

so (a_n) cauchy. now, check $p_k \to [(a_n)]$. let $\varepsilon > 0$, choose N s.t. $d(a_n, a_m) < \varepsilon \ \forall n, m \geq N$

and $1/N < \varepsilon$. for $j \geq N$:

$$\begin{aligned} D(p_j, [(a_n)]) &= \lim_{n\to\infty} d(p_n^{(j)}, p_m^{(j)}) \\ &\leq \limsup_{n\to\infty} d(p_n^{(j)}, p_j^{(j)}) \ + \ d(p_j^{(j)}, p_m^{(j)}) \\ &\leq 1/j + \limsup_{n\to\infty} d(a_j, a_n) \\ &\leq 2\varepsilon \end{aligned}$$

so M' is complete. [verify, cant see that board v well]

pf that M' dense: exercise: if $p([(p_n)])$, want to find $(a_n) \in M$ s.t. $\iota(a_n) \to p$ in M' [what]. set $a_n = p_n$. show $p_n = [(a_n, a_n, \dots)]$ and show $D(p_n, p) \to 0$. [what]

proof #2 of every space having a completion. bro can we move tf on. suffices to show $\exists$ complete X and map $\iota : M \longrightarrow X$ which is isometry onto img. then, take $M' = \overline{\iota(M)}$.

let $X = C_0(M)$ be space of cts $\mathbb{R}$-valued fn's on M under sup metric. we showed $C([0,1])$ complete - same pf shows X complete. we show ι exists.

fix $x_0 \in M$. for any $x \in M$, defn $f_x : M \longrightarrow \mathbb{R}$ by:

$$f_x(y) \ := \ d(x,y) - d(x_0, y)$$

clearly, f_x cts. also bdd since by rev. triangle:

$$|d(x,y) - d(x_0,y)| \ \leq \ d(x, x_0)$$

claim: $x \to f_x$ an isometry. pf:

$$|f_x(y) - f_z(y)| \ = \ |d(x,y) - d(z,y)| \ \leq \ d(x,z)$$

note equality is at $y = x$ or z. thus:

$$d(f_x, f_z) \ = \ \sup_{y\in M} |f_x(y) - f_z(y)| \ = \ d(x,z)$$

caveat w this pf: requires $\mathbb{R}$ being complete.

defn: dense, separable.

Lemma 2.12

M cpt implies separable.

Proof. Source: Primary Source Material

for $\varepsilon = 1/n$, $M \subseteq \bigcup_M B(x, 1/n)$. this has finite subcvr, so:

$$M \subseteq \bigcup_{j=1}^{k_n} B(x_j^{(??)}, 1/n)$$

define:

$$S := \bigcup_{n=1}^{\infty} \left\{ x_1^{(n)}, \ldots, x_{k_n}^{(n)} \right\}$$

easy to check S is dense. [where tf did he pull this out of] ■

II Metric Spaces of Functions

3 Basics

ugh finally

Definition 3.1

given metric (X, d), we defn:

$$B(X) \ := \ \{\text{bounded real-valued fn's on } X\}$$

and $C_0(X) \subseteq B(X)$ as the cts ones. we defn the metric as:

$$D(f, g) \ := \ \|f - g\|_\infty \ = \ \sup_{z \in X} |f(z) - g(z)|$$

note the subscripts - C_0 means bdd, C^0 not necessarily.

Theorem 3.2

$B(X), C_0(X)$ are complete.

Proof.

same as for $C([0, 1])$. let $f_n \in B(X)$ be cauchy. $\forall z$:

$$|f_n(z) - f_m(z)| \leq \|f_n - f_m\|_\infty$$

so $\{f_n(z)\}$ cauchy in $\mathbb{R}$. set $f(z) := \lim_n f_n(z)$. need to check $f_n \to f$, f bdd, and f cts if $f_n \in C_0(X)$.

let $\varepsilon > 0$. then $\exists N$ s.t. $\|f_n - f_m\|_\infty < \varepsilon$ for $n, m \geq N$. for $n \geq N$:

$$|f(z) - f_n(z)| \ = \ \lim_m |f_m(z) - f_n(z)| \ \leq \ \limsup_m \|f_m - f_n\|_\infty \leq \varepsilon$$

so $\|f - f_n\| \to 0$.

for bdd, apply above w $\varepsilon = 1$. then

$$|f(z)| \;\leq\; |f(z) - f_N(z)| + |f_N(z)| \;\leq\; \|f - f_N\|_\infty + \|f_N\|_\infty \;\leq\; 1 + \|f_N\|_\infty$$

so f bdd. for $C_0(X)$, do the usual $\varepsilon/3$ argument. ■

in particular, this shows $C_0(X)$ closed in $B(X)$.

Definition 3.3

given $f_n : X \longrightarrow \mathbb{R}$, we say $f_n \to f$ (ptwise) if:

$$\lim_{n\to\infty} f_n(z) = f(z)$$

for all z. we say $f_n \to f$ **uniformly** if:

$$\lim_{n\to\infty} \sup_z |f_n(z) - f(z)| = 0 \iff \lim_{n\to\infty} \|f_n - f\|_\infty = 0$$

Lemma 3.4

let $f_n : [0,1] \longrightarrow \mathbb{R}$ be seq of riemann int'bl fn's s.t. $f_n \to f$ uniformly.

(i) f riemann int'bl

(ii) ints and lims are exchangeable, i.e.

$$\lim_{n\to\infty} \int_0^1 f_n(x)\mathrm{d}x \;=\; \int_0^1 \lim_{n\to\infty} f_n(x)\mathrm{d}x \;=\; \int_0^1 f(x)\mathrm{d}x$$

Proof. Source: Primary Source Material

(i) see mat157 [he acc wrote this]

(ii) notice that:

$$\left|\int_0^1 f_n(x) - f(x)\mathrm{d}x\right| \leq \int_0^1 |f_n(x) - f(x)|\mathrm{d}x \leq \|f_n - f\|_\infty \to 0$$

as $n \to \infty$. ■

Lemma 3.5

let $f_n : (0,1) \longrightarrow \mathbb{R}$. sps:

(i) $f_n \to f$ uniformly

(ii) f_n' exists

(iii) $f_n' \to g$ uniformly

then f diff'ble and $f' = g$.

Proof. Source: Primary Source Material

fix $t \in (0,1)$. consider:

$$\phi(x) = \begin{cases} \frac{f(x)-f(t)}{x-t} & x \neq t \\ g(t) & x = t \end{cases}$$

suffices to check ϕ cts. consider:

$$\phi_n(x) = \begin{cases} \frac{f_n(x)-f_n(t)}{x-t} & x \neq t \\ f_n'(t) & x = t \end{cases}$$

note $\phi_n(x) \to \phi(x)$ ptwise. wts ϕ_n uni cvgs; can check ϕ_n cauchy in $C_0([0,1])$.

if $x = t$, then:

$$|\phi_n(t) - \phi_m(t)| \;=\; |f_n'(t) - f_m'(t)| \;\leq\; \|f_n' - f_m'\|_\infty$$

if $x \neq t$, then:

$$|\phi_n(x) - \phi_m(x)| \;=\; \left|\frac{(f_n(x) - f_m(x)) - (f_n(t) - f_m(t))}{x - t}\right|$$

denote $F_{nm}(x) := f_n(x) - f_m(x)$. by mvt, the above is equal to:

$$|\phi_n(x) - \phi_m(x)| \;=\; |F_{nm}'(\theta)| \;=\; |f_n'(\theta) - f_m'(\theta)| \;\leq\; \|f_n' - f_m'\|_\infty$$

for some $\theta \in (0, 1)$. thus $\|\phi_n - \phi_m\|_\infty \leq \|f_n' - f_m'\|_\infty$, so cvg is uniform. ■

Example 3.6 Source: Primary Source Material

consider $C_0([0, 1])$ and $f_n(x) = x^n$. bdd w $\|f_n\|_\infty \leq 1$, but no cvg subseq.

Definition 3.7

let X be metric sp, $F \subseteq C_0(X)$. we say F is **equicontinuous** if $\forall \varepsilon > 0$, $\exists \delta > 0$ s.t. $|f(x) - f(y)| < \varepsilon$ $\forall x, y$ satisfying $d(x, y) < \delta$ and *all* $f \in F$.

note: if F singleton, this means that particular fn is uni cts.

Example 3.8 Source: Primary Source Material

the set $F_K = \{f \in C_0([0, 1]) : f' \text{ exists and } \|f'\|_\infty \leq K\}$. is equicts. if $f \in F_K$:

$$|f(x) - f(y)| \;=\; |f'(s)||x - y| \;\leq\; K|x - y|$$

for some $s \in (0, 1)$. note, however, that $F = \bigcup_K F_K$ *not* equicts.

4 Arzela-Ascoli Theorem

Lec 9 - Feb 3 (Week 5)

Theorem 4.1: Arzela-Ascoli

let X be cpt, $(f_n)_n$ seq of fn's in $C_0(X)$ s.t.:

(i) $\exists C > 0$ s.t. $\|f_n\|_\infty < C$

(ii) $F = \{f_n\}$ equicts

then $\exists$ subseq of f_n which uni cvgs to some $f \in C_0(X)$.

prob wont put whole pf in a box, we'll see.

Lemma 4.2

sps E ctbl, $f_n : E \longrightarrow \mathbb{R}$ uni bdd, i.e.:

$$\sup_n \sup_{x \in E} |f_n(x)| < \infty$$

then $\exists$ subseq f_{n_k} s.t. $\lim_{k\to\infty} f_{n_k}(x)$ exists $\forall x \in E$.

Proof. Source: Primary Source Material

sps $E = \{x^{(1)}, \dots\}$ and defn $a_n^{(k)} := f_n(x^{(k)})$. notice $(a_n^{(1)})$ a bdd seq of real nums, so $\exists$ subseq:

$$S_1 := (n_1^{(1)}, n_2^{(1)}, \dots) \qquad \lim_{j\to\infty} a^{(1)}_{n_j^{(1)}} \text{ exists}$$

then for all $k \geq 2$, we can find subseq $S_k := (n_1^{(k)}, n_2^{(k)}, \dots)$ such that:

- S_k subseq of S_{k-1}
- $\lim_{j\to\infty} a^{(k)}_{n_j^{(k)}}$ exists

each S_k constructed by induction [in the obv way].

now, let $m_k := n_k^{(k)}$. claim: (m_k) a subseq, i.e. $m_{k+1} > m_k$, and $\lim_k f_{m_k}(x)$ exists for all $x \in E$. [note - don't forget we're working on indices]

note m_j an el't of k-th subseq S_k for $j \geq k$. thus, for any k:

$$\lim_{j\to\infty} f_{m_j}(x^{(k)}) \;=\; \lim_{j\to\infty} f_{n_j^{(k)}}(x^{(k)})$$

exists by construction of S_k. ■

pf of arzela-ascoli.

since X cpt, X separable by some E. by lemma, $\exists$ subseq (n_k) s.t. $\forall x \in E$:

$$\lim_{k\to\infty} f_{n_k}(x) < \infty$$

thus, suffices to show f_{n_k} is cauchy, i.e. $\left\|f_{n_j} - f_{n_k}\right\|_\infty < \varepsilon$.

let $\varepsilon > 0$. since F equicts, $\exists \delta > 0$ s.t. $|f_n(x) - f_n(y)| < \varepsilon$ for all x, y s.t. $d(x,y) < \delta$ and all n. since E dense, $X = \bigcup_E B(x,\delta) = \bigcup_{i=1}^m B(x_i,\delta)$ since X cpt.

now, $(f_{n_k}(x_i))$ cauchy $\forall i$, so [for suff. large k, j], $\left|f_{n_k}(x_i) - f_{n_j}(x_i)\right| < \varepsilon$. let $x \in X$, $k, j > N$. then $x \in B(x_i, \delta)$ for some i. then:

$$\left|f_{n_k}(x) - f_{n_j}(x)\right| \;\leq\; |f_{n_k}(x) - f_{n_k}(x_i)| + \left|f_{n_k}(x_i) - f_{n_j}(x_i)\right| + \left|f_{n_j}(x_i) - f_{n_j}(x)\right| \;\leq\; 3\varepsilon$$
$$\implies \left\|f_{n_k} - f_{n_j}\right\|_\infty \leq 3\varepsilon$$

so the seq is cauchy.

Theorem 4.3

sps X cpt. a set $F \subseteq C_0(X)$ cpt iff closed, bdd, equicts.

Proof. Source: Primary Source Material

"if" dir: see AA.

"only if" dir: if F cpt, its closed, bdd, totally bdd. thus $\forall \varepsilon > 0$:

$$F \subseteq \bigcup_{j=1}^{n} B(f_j, \varepsilon)$$

since X cpt, (f_j) uni cts. thus $\exists \delta > 0$ s.t. $|f_j(x) - f_j(y)| < \varepsilon$ for x, y such that $d(x, y) < \delta$ and all f_j.

now, let $x, y \in X$ s.t. $d(x, y) < \delta$, $f \in F$. then $\exists f_j$ s.t. $f \in B(f_j, \varepsilon)$, so:

$$|f(x) - f(y)| \leq |f(x) - f_j(x)| + |f_j(x) - f_j(y)| + |f_j(y) - f(y)| \leq 3\varepsilon$$

and we're done. ■

5 Convolutions and Weierstrass Approximation

(Midterm last thurs) Lec 10 - Feb 10 (Week 6)

we say a fn is cptly supported if its supp is cpt.

Definition 5.1

if $f, g : \mathbb{R} \longrightarrow \mathbb{R}$ cptly supp and riemann int'bl, then:

$$(f * g)(x) = \int_{\mathbb{R}} f(x - t) g(t) \mathrm{d}t$$

is well defn'd, and is called the **convolution** of f and g.

Definition 5.2

a seq $J_n : \mathbb{R} \longrightarrow \mathbb{R}$ of cptly supp riemann int'bl fn's are called an **approximate identity** if:

(i) $\displaystyle\int_{\mathbb{R}} I_n(x)\mathrm{d}x = 1$

(ii) $\sup_n \displaystyle\int_{\mathbb{R}} |I_n(x)|\mathrm{d}x < \infty$

(iii) $\displaystyle\lim_{n\to\infty} \int_{|x|>\delta} |I_n(x)|\mathrm{d}x = 0$ for all $\delta > 0$

Theorem 5.3

let $f : [a,b] \longrightarrow \mathbb{R}$ be cts, $f(a) = f(b) = 0$. define $\widetilde{f}(x)$ as:

$$\widetilde{f}(x) \;=\; \begin{cases} f(x) & x \in [a,b] \\ 0 & \text{otw} \end{cases}$$

let I_n be an approx id. then $f_n = \widetilde{f} * I_n$ cvgs uniformly to $\widetilde{f}$.

Proof. Source: Primary Source Material

let $\varepsilon > 0$. easy to see: $\widetilde{f}$ uni cts, so $\exists \delta > 0$ s.t.

$$\left|\widetilde{f}(x) - \widetilde{f}(y)\right| < \varepsilon \qquad \forall |x-y| < \delta$$

then:

$$\begin{aligned} f_n(x) - \widetilde{f}(x) &= \int \widetilde{f}(x-y) I_n(y) \mathrm{d}y - \int \widetilde{f}(x) I_n(y) \mathrm{d}y \\ &= \int_{\mathbb{R}} (\widetilde{f}(x-y) - \widetilde{f}(x)) I_n(y) \mathrm{d}y \\ &= \int_{|y|<\delta} (\widetilde{f}(x-y) - \widetilde{f}(x)) I_n(y) \mathrm{d}y + \int_{|y|>\delta} (\widetilde{f}(x-y) - \widetilde{f}(x)) I_n(y) \mathrm{d}y \end{aligned}$$

first term:

$$\leq \int_{|y|<\delta} \left| \widetilde{f}(x-y) - \widetilde{f}(x) \right| |I_n(y)| \mathrm{d}y \leq \varepsilon \cdot \sup_n \int_{\mathbb{R}} |I_n(y)| \mathrm{d}y$$

2nd term:

$$\leq 2 \left\| \widetilde{f} \right\|_\infty \int_{|y|>\delta} |I_n(y)| \mathrm{d}y$$

goes to 0 as $n \to \infty$. thus, $\limsup_{n\to\infty} \sup_{x\in\mathbb{R}} \left| f_n(x) - \widetilde{f}(x) \right| \leq C\varepsilon$. ■

Theorem 5.4: Weierstrass Approximation Theorem

sps $f : [a, b] \longrightarrow \mathbb{R}$ cts. then $\exists$ seq of polyn's p_n s.t. $p_n \rightrightarrows f$ for $x \in [a, b]$.

Proof. Source: Primary Source Material

by subtracting linear fn, wlog $f(a) = f(b) = 0$. by rescaling, wlog $a = 0$ and $b = 1$.

extend f to $\mathbb{R}$ with:

$$\widetilde{f}(x) = \begin{cases} f(x) & x \in [0, 1] \\ 0 & \text{otw} \end{cases}$$

define $\beta_n(t) := b_n(1 - t^2)^n \mathbb{I}_{|t| \leq 1}$, where $\mathbb{I}$ indicator fn, and:

$$b_n = \frac{1}{\int_{-1}^{1} (1 - t^2)^n \mathrm{d}t}$$

we claim the following:

(1) β_n an approx id

(2) $f_n(x) = (\widetilde{f} * \beta_n)(x)$ a polyn $\forall n$ for $x \in [0, 1]$

together, these prove the statement.

first, we prove claim 2. we have:

$$f_n(x) = \int_{-1}^{1} \widetilde{f}(x - t)\beta_n(t)\mathrm{d}t$$

by change of variables $u = x - t$, this is equal to:

$$\int_{x-1}^{x+1} \widetilde{f}(u)\beta_n(x - u)\mathrm{d}u = \int_{0}^{1} f(u)\beta_n(x - u)\mathrm{d}u$$

if $x \in [0,1]$ and $u \in [0,1]$, then $x - u \in [-1,1]$, so this is equal to:

$$b_n \int_0^1 f(u)(1-(x-u)^2)^n \mathrm{d}u$$

this is a polyn in x, since this is equal to:

$$\sum_{k,j} x^j \int_0^1 f(u) u^k \mathrm{d}u$$

now, we prove claim 1. by defn:

$$\int_{\mathbb{R}} \beta_n(t)\mathrm{d}t \;=\; \int_{\mathbb{R}} |\beta_n(t)|\mathrm{d}t \;=\; 1$$

now, we show:

$$\int_{|t|>\delta} \beta_n(t)\mathrm{d}t \;\to\; 0 \text{ as } n \to \infty$$

we find an upper bound for b_n:

$$\begin{aligned}
\int_{-1}^1 (1-t^2)^n \mathrm{d}t &\geq \int_{-1/\sqrt{n}}^{1/\sqrt{n}} (1-t^2)^n \mathrm{d}t \\
&\geq \int_{-1/\sqrt{n}}^{1/\sqrt{n}} \left(1 - \left(\frac{1}{\sqrt{n}}\right)^2\right)^n \mathrm{d}t \\
&= \frac{2}{\sqrt{n}} \left(1 - \frac{1}{n}\right)^n \\
&\geq \frac{C}{\sqrt{n}}
\end{aligned}$$

holds for all $n \geq 2$. thus $b_n \leq C\sqrt{n}$, so:

$$\begin{aligned}\int_{|t|>\delta} \beta_n(t)\mathrm{d}t &\leq C\sqrt{n}\int_{|t|>\delta}(1-t^2)^n \mathbb{I}_{|t|\leq 1} \\ &\leq C\sqrt{n}(1-\delta^2)^n\end{aligned}$$

this goes to 0 as $n \to \infty$ as needed.

we have proved both claims, thus the statement is proved. ■

6 Equicontinuity and Stone-Weierstrass

why are we defining algebras now lmfao

Definition 6.1

we say $A \subseteq C_0(M)$ **separates pts** if $\forall p, q \in M$, $\exists f \in A$ s.t. $f(p) \neq f(q)$.

we say A **vanishes** at $x \in M$ if $f(x) = 0$ $\forall f \in A$.

Theorem 6.2: Stone-Weierstrass

if M cpt, $A \subseteq C_0(M)$ an alg of fn's which separates pts and vanishes nowhere, then $\overline{A} = C_0(M)$.

Lec 11 - Feb 12 (Week 6)

before we prove stone-weierstrass, we need the following lemma.

Lemma 6.3

if $A \subseteq C_0(M)$ an alg, then so is $\overline{A}$.

Proof. Source: Primary Source Material

we'll just prove $\overline{A}$ closed under products. let $f, g \in \overline{A}$.

then $\exists f_n, g_n \in A$ s.t. $f_n \rightrightarrows f$ and $g_n \rightrightarrows g$. then $f_n g_n \rightrightarrows fg$, so $fg \in \overline{A}$. ■

Lemma 6.4

if $A \subseteq C_0(M)$ an alg which separates pts + vanishes nowhere, then $\forall p_1 \neq p_2 \in M$ and $c_1, c_2 \in \mathbb{R}$, $\exists H \in A$ s.t. $H(p_1) = c_1$ and $H(p_2) = c_2$.

Proof. Source: Primary Source Material

since A vanishes nowhere, $\exists g_1, g_2$ s.t. $g_1(p_1) \neq 0$ and $g_2(p_2) \neq 0$. then $g = g_1^2 + g_2^2$ doesn't vanish at either p_i.

let $f \in A$ s.t. $f(p_1) \neq f(p_2)$. consider:

$$M = \begin{pmatrix} g(p_1) & f(p_1)g(p_1) \\ g(p_2) & f(p_2)g(p_2) \end{pmatrix} \qquad \det(M) = g(p_1)g(p_2)(f(p_2) - f(p_1)) \neq 0$$

therefore, the eq'n:

$$M \begin{pmatrix} \xi \\ \eta \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

has a soln (ξ, η). thus, $H = \xi g + \eta f g \in A$ satisfies $H(p_1) = c_1, H(p_2) = c_2$. ■

we now prove stone-weierstrass. dont feel like boxing it

first, notice if $f \in \overline{A}$, then $|f| \in \overline{A}$. to see this, suffices to find $f_n \in \overline{A}$ s.t. $f_n \rightrightarrows |f|$. consider:

$$I = [-\|f\|_\infty, \|f\|_\infty]$$

by weierstrass approx. thm, $\exists p_n(x)$ polys s.t. $p_n \rightrightarrows |x|$ on I. consider $f_n = p_n(f)$. since $\overline{A}$ an alg, $f_n \in \overline{A}$ for all n. on the other hand, since $p_n \rightrightarrows |x|$ on I, then $f_n \rightrightarrows |f|$ on M.

next, if $f, g \in \overline{A}$, then:

$$\max(f,g) \;=\; \frac{f+g}{2} + \frac{|f-g|}{2} \;\in\; \overline{A} \qquad \min(f,g) \;=\; \frac{f+g}{2} - \frac{|f-g|}{2} \;\in\; \overline{A}$$

finally, max/min of finitely many elems of $\overline{A}$ are also in $\overline{A}$. we now prove the statement. given $F \in C_0(M)$ and $\varepsilon > 0$, want to find $G \in \overline{A}$ s.t. $\forall x \in M$:

$$F(x) - \varepsilon \;\leq\; G(x) \;\leq\; F(x) + \varepsilon$$

by lemma, $\forall p, q \in M$, $\exists H_{pq} \in \overline{A}$ s.t.:

$$H_{pq}(p) = F(p) \qquad H_{pq}(q) = F(q)$$

fix $p \in M$. then $\forall q \in M$, $\exists U_q \ni q$ nbhd s.t. $\forall x \in U_q$:

$$H_{pq}(x) \;>\; F(x) - \varepsilon$$

now, $\{U_q\}$ set of nbhds arnd every pt q forms an open cvr of M. since M cpt, then $\exists \{U_{q_1}, \ldots, U_{q_n}\}$ cvring M. consider:

$$G_p(x) \;=\; \max_j H_{pq_j}(x)$$

note $G_p \in \overline{A}$, and $G_p(x) > F(x) - \varepsilon$ for all $x \in M$. the first inequality has been satisfied. now, $\forall p \in M$, $\exists$ nbhd $V_p \ni p$ s.t. $\forall x \in V_p$:

$$G_p(x) < F(x) + \varepsilon$$

as before, $V_{p_1}, \ldots, V_{p_m}$ cvr M. let:

$$G \;=\; \min_j G_{p_j} \in \overline{A}$$

similar to before, we get that $G(x) < F(x) + \varepsilon$ for all M. OTOH, $G(x) > F(x) - \varepsilon$ for all M, so we have found our desired function. $\square$

we need to patch a small hole near the start: in particular, $\overline{A}$ may not contain any constant fn's.

Lemma 6.5

sps $I = [-M, M]$. $\exists$ seq of polys w/ no constant term s.t. $p_n \rightrightarrows |x|$ in I.

Proof. Source: Primary Source Material

by weierstrass approx. thm, $\exists q_n(x) = a_n + p_n(x)$ s.t. $q_n \rightrightarrows |x|$. then:

$$q_n(0) \to |0| = 0 \implies a_n = q_n(0) \to 0 \text{ as } n \to \infty$$

since $a_n \to 0$, then $p_n = q_n - a_n \rightrightarrows |x|$. ■

preview of a later topic

Definition 6.6

we say f is a **trigonometric polynomial** if:

$$f(x) = a_0 + \sum_{k=1}^{n} a_k \cos(kx) + \sum_{k=1}^{m} b_k \sin(kx)$$

as finite sums.

Theorem 6.7

let f be 2π-periodic cts. then $\exists p_n$ seq of trig polys s.t. $p_n \rightrightarrows f$.

want to use stone-weierstrass, so want to restrict to $[0, 2\pi]$. however, 0 and 2π are not separated, and we can't exclude either pt while maintaining cptness. idea: we identify them together instead.

Proof. Source: Primary Source Material

Identify the space of 2π-periodic cts $f : \mathbb{R} \longrightarrow \mathbb{R}$ w cts $\widetilde{f} : S^1 \longrightarrow \mathbb{R}$ by setting $\widetilde{f}(e^{i\theta}) = f(\theta)$ for $\theta \in [0, 2\pi]$.

need to check trig polys satisfy hypotheses as elems of $C_0(S^1)$. indeed, they're an alg bc of trig identities, nowhere vanishing b/c they include constant fn's, and they separate pts (exercise).

thus, by stone-weierstrass, we're done. ■

7 Two Metric Space Theorems

r we rly doing banach fixed pt lol. hm. weird defn of a contraction. oh i lied its standard

Definition 7.1

$f : M \longrightarrow M$ a **contraction** if $\exists \alpha \in [0, 1)$ s.t. $d(f(x), f(y)) \leq \alpha d(x, y)$ on all M.

statement of banach fixed pt for completion

Theorem 7.2: Banach Fixed-Point Theorem

sps M complete, $f : M \longrightarrow M$ a contraction. then $\exists! x \in M$ s.t. $f(x) = x$.

Example 7.3 Source: Primary Source Material

let (a_{ij}) be doubly indexed real nums s.t. $\sup_{i,j \geq 1} |a_{ij}| \leq K$ and $\sum_{i,j} |a_{ij}| < \infty$ for some $K > 0$. let $b \in \ell^1$ and consider:

$$F : \ell^1 \longrightarrow \ell^1 \qquad (F(x))_i = \sum_j a_{ij} x_j^2 + b_j$$

then:

$$\sum_i (F(x))_i \;\leq\; \sum_i \left(\sum_j |a_{ij}||x_j|^2 + |b_i| \right) \;\leq\; \|x\|_\infty^2 \sum_{ij} |a_{ij}| + |b|_1 < \infty$$

we have $F : \ell^1 \longrightarrow \ell^1$ is well-defn'd. we show that $\exists \varepsilon > 0$ s.t. if $\|b\|_1 \leq \varepsilon$, then $\exists x \in \ell^1$ s.t. $x = F(x)$.

idea: wtfind closed set $\subseteq \ell^1$ on which F contracts. let $x, y \in \ell^1$. then:

$$\begin{aligned}
\|F(x) - F(y)\|_1 \;&\leq\; \sum_i \left| \sum_j a_{ij}(x_j^2 - y_j^2) \right| \\
&\leq\; \sum_i \sum_j |a_{ij}| \left(|x_j| + |y_j| \right) |x_j - y_j| \\
&\leq\; \|x - y\|_\infty (\|x\|_\infty + \|y\|_\infty) \left(\sum_{i,j} |a_{ij}| \right) \\
&\leq\; \|x - y\|_1 (\|x\|_1 + \|y\|_1) M
\end{aligned}$$

note that as long as $x, y \in \overline{B}(0, r)$ with $r = 1/(3M)$, then F is a contraction. lastly, need to check that $F(\overline{B}(0, r)) \subseteq \overline{B}(0, r)$.

$$\begin{aligned}
\|F(x)\|_1 \;&\leq\; \sum_i \left| \sum_j a_{ij} x^2 + b_i \right| \\
&\leq\; \|x\|_\infty^2 \sum_{i,j} |a_{ij}| + \|b\|_1 \\
&\leq\; \|x\|_1^2 M + \|b\|_1 \\
&\leq\; \frac{M}{9M^2} + \varepsilon
\end{aligned}$$

whered that last bit come from. what. anyway as long as $M \geq 1$ and $\varepsilon \leq 1/(9M)$:

$$\|F(x)\|_1 \leq \frac{1}{9M} + \frac{1}{9M} \leq \frac{1}{3M}$$

so we can apply banach fixed pt.

point of the example: given non-linear fn's on a general space(?), it may not be a contraction on the whole space. however, we might be able to find a suff. small closed ball on which it is in fact a contraction; if the fn also maps the ball to itself, then we're good.

Lec 12 - Feb 24 (Week 7)

motivation: what does it mean for a pt in a space to be "generic"? for instance, in $\mathbb{R}$, most pts are in some sense, probably irrational. note:

$$\overline{\mathbb{Q}} = \mathbb{R} \qquad \overline{\mathbb{Q}^c} = \mathbb{R} \qquad \overline{\mathbb{Q} \cap \mathbb{Q}^c} = \varnothing$$

so density alone isnt a good test.

Lemma 7.4

let $A, B \subseteq M$ be open dense. then $A \cap B$ is open dense.

Proof. Source: Primary Source Material

fix $x \in M$ and $\varepsilon > 0$. since A dense, $\exists a \in A$ s.t. $a \in B(x, \varepsilon)$. since $B(x, \varepsilon) \cap A$ open, $\exists r > 0$ s.t. $B(a, r) \subseteq B(x, \varepsilon) \cap A$. since B dense, $\exists b \in B(a, r)$. then $b \in A \cap B \cap B(x, \varepsilon)$ as needed. ■

we can now state bct.

Theorem 7.5: Baire Category Theorem

sps M complete. let G_n be a seq of open dense subsets of M. then:

$$G = \bigcap_{n=1}^{\infty} G_n$$

is dense.

sets of this form have many names, such as:

- thick.
- dense G_δ sets, i.e. a ctbl intersection of open sets

Corollary 7.6

if M complete and:

$$M = \bigcup_{n=1}^{\infty} F_n$$

where F_n closed, then at least one F_n has non-empty interior.

Proof. Source: Primary Source Material

if M is as above, then $\varnothing = \bigcap_n F_n^c$. exercise: F_n^c is dense iff F_n has no interior. ■

we now prove bct.

Proof. Source: Primary Source Material

let $x \in M$, $\varepsilon > 0$. wts $\exists p \in G \cap B(x, \varepsilon)$. we'll construct a cauchy seq inductively, w pts p_n and radii r_n s.t. $B(p_1, 2r_1) \subseteq B(x, \varepsilon)$, and:

$$B(p_n, 2r_n) \subseteq B(p_{n-1}, r_{n-1}) \cap \bigcap_{k=1}^{n-1} G_k \ =: \ B$$

to start, pick $p_1 = x$ and $r_1 = \varepsilon/2$. now, for any n, $\bigcap_{k=1}^{n-1} G_k$ open dense. thus, $\exists y \in B$. since B is open, $\exists r > 0$ s.t. $B(y, r) \subseteq B$. thus, take $p_n = y$ and $r_n = \min(r/2, 1/n)$. since $r_n \to 0$, p_n is cauchy, thus $p_n \to p$. since $p_k \in B(p_n, r_n)$ for all $k \geq n$, we have:

$$p \in \overline{B(p_n, r_n)} \subseteq B(p_n, 2r_n)$$

since $p \in B(p_n, 2r_n)$ for all n, then clearly $p \in B(x, \varepsilon)$ and $p \in G_n \ \forall n$ as needed. ■

kinda missed what he said but sth abt ill-behaved fn's.

Proposition 7.7

$\exists$ a thick $A \subseteq C_0([a, b])$ s.t. $\forall f \in A$:

1. the left/right derivs dne for any $x \in [a, b]$
2. f is not monotone on any $I = (c, d)$.

Proof. Source: Primary Source Material

define:

$$\Delta_h f(x) \ := \ \frac{f(x+h) - f(x)}{h}$$

$$\begin{aligned} R_n \ &:= \ \{f : \forall x \in [a, b - 1/n], \exists h < 1/n, |\Delta_h f_x| > n\} \\ L_n \ &:= \ \{f : \forall x \in [a + 1/n, b], \exists -1/n < h < 0, |\Delta_h f_x| > n\} \\ G_n \ &:= \ \{f : f \text{ not monotone on any interval } I \subseteq [a, b] \text{ of length } 1/n\} \end{aligned}$$

need to check:

i) R_n, L_n, G_n open + dense

ii) $\bigcap_{n=1}^{\infty} R_n \cap L_n \cap G_n$ has the required properties

second pt is clear; we check the first pt. to see R_n open, note:

$$|\Delta_h f(x) - \Delta_h g(x)| \leq \frac{2\|f-g\|_\infty}{h}$$

let $f \in R_n, x \in [a, b-1/n]$. then $\exists h_x$ s.t. $|\Delta_{h_x} f(x)| > n$. since f cts, $\exists T_x \ni x$ open intvl and $v_x > 0$ s.t. $\forall t \in T_x$:

$$|\Delta_h f(t)| > n + v_x$$

in particular, by cptness:

$$[a, b-1/n] \;=\; \bigcup_{x \in [a,b-1/n]} T_x \;=\; \bigcup_{j=1}^{m} T_{x_j}$$

let $h = \min(h_{x_j})$ and $v = \min(v_{x_j})$. then, if $\varepsilon < vh/2$ and $\|g-f\|_\infty < \varepsilon$, we have:

$$\left|\Delta_{h_{x_j}} g(t)\right| \;>\; \left|\Delta_{h_{x_j}} f(t)\right| - \frac{2\varepsilon}{h_{x_j}} \;>\; n + v_{x_j} - \frac{vh}{h_{x_j}} \;>\; n + v - v \;=\; n$$

for $t \in T_{x_j}$. in particular, $g \in R_n$, so R_n and similarly L_n open. pf cont below. ■

(skipped) Lec 13 - Feb 26 (Week 7)

we now check that G_n is open, or G_n^c closed. let $f_m \in G_n^c$ s.t. $f_m \to f$. then $\exists I_m = [c_m, d_m]$ with $d_m - c_m = 1/n$ s.t. f_m is monotone on I_m. by taking a subseq, sps wlog $c_m \to c$ and $d_m \to d$, and all f_m either increasing or decreasing. then since $f_m \to f$ ptwise, f monotone on $[a, b]$, so $f \in G_n^c$. thus, G_n^c closed, so G_n open.

it remains to see that $R_n \cap L_n \cap G_n$ dense. by weierstrass approx, it suffices to show that poly's are in the closure. indeed, let f be a poly. let $\theta_m : \mathbb{R} \longrightarrow \mathbb{R}$ be the fn s.t.

$\theta_m(k/m)$ is 0 for even k and 1 for odd k, and linear btwn them (draw a picture). let $\varepsilon > 0$, $f_m = f + \varepsilon\theta_m/2$. then $\|f - f_m\|_\infty < \varepsilon$, so it suffices to check $f_m \in R_n \cap L_n \cap G_n$ for suff. large m. for any x, h, we have:

$$|\Delta_h f_m(x)| \;\geq\; \frac{\varepsilon}{2}|\Delta_h \theta_m(x)| - \sup_{x\in[a,b]} |f(x)|a \;\geq\; \frac{\varepsilon}{2}|\Delta_h \theta_m(x)| - C$$

for some $C > 0$ indep of h, x. then for $x \in \mathbb{R}$, note (or check) $\exists 0 < h < 2/m$ s.t.:

$$|\Delta_h \theta_m(x)| > \frac{m}{4}$$

thus for suff. large m, $f_m \in R_n$ and similarly L_n. to see that $f_m \in G_n$ for large m, note that for any interval I of len $1/n$, if $m > 10n$, $\exists x < y < z$ s.t. $|x - z| < 1/m$ and:

$$\theta_m(y) \geq \theta_m(x) + 1 \qquad \theta_m(z) \leq \theta_m(y) - 1$$

OTOH, $|f(y) - f(x)| \leq \|f'\|_\infty/m$ and $|f(z) - f(y)| \leq \|f'\|_\infty/m$, so for suff. large m:

$$f_m(y) \geq f_m(x) + \frac{\varepsilon}{4} \qquad f_m(z) \leq f_m(y) - \frac{\varepsilon}{4}$$

and the result follows.

8 Fourier Series

recall that a trig poly is a fn of the form:

$$f(x) \;=\; b_0 + \sum_{k=1}^{N} b_k \cos(kx) + \sum_{k=1}^{N} a_k \sin(kx)$$

we previously showed these can approx 2π-periodic cts fn's, but this was non-constructive. now, we aim for an explicit approximation. we'll modify our setup and consider:

$$f(x) \;=\; \sum_{k=-N}^{N} c_k e^{ikx}$$

where $c_k \in \mathbb{C}$. if a_k, b_k are complex, these forms become equivalent. furthermore, f is real iff $c_k = \overline{c_{-k}}$ (i think? his notations weird). recall (sure man):

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{ikx}\ \mathrm{d}x \ = \ \begin{cases} 1 & k = 0 \\ 0 & k \neq 0 \end{cases}$$

thus, if f is of the above form, then:

$$c_k \ = \ \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-ikx}\ \mathrm{d}x$$

in particular, for any riemann int'bl $f : [-\pi, \pi] \longrightarrow \mathbb{C}$, we defn:

$$\widehat{f}(k) \ := \ c_k = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-ikx}\ \mathrm{d}x$$

and ask whether the series:

$$\sum_k c_k e^{ikx}$$

cvgs to f in some sense. this is the content of this section.

Definition 8.1

inner product lol. map $\langle \cdot, \cdot \rangle : V \times V \longrightarrow \mathbb{C}$ with:

i) $\langle u, v \rangle = \overline{\langle v, u \rangle}$

ii) $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$

iii) $\langle u, \alpha v \rangle = \langle \overline{\alpha} u, v \rangle = \alpha \langle u, v \rangle$

iv) $\langle v, v \rangle \geq 0$ w equality iff $v = 0$

this defns a norm $\|f\| = \langle f, f \rangle^{1/2}$. taking $V = C[a, b]$ over *complex* fn's, we consider:

$$\langle f, g \rangle \ := \ \int_a^b \overline{f(x)} g(x)\ \mathrm{d}x$$

this is *almost* an inn prod, but fails iv) (idt it does..? but sure). however, thats not important for what we want. we'll henceforth call $\|f\| = \langle f, f\rangle^{1/2}$ for this section.

Definition 8.2

a seq of riemann int'bl fn's $\psi_n : [a,b] \longrightarrow \mathbb{C}$ is called **orthogonal** if ψ_n, ψ_m are ortho for $n \neq m$. we say they are **orthonormal** if... well yk.

eg, taking $[a,b] = [-\pi, \pi]$, $\phi_n = \frac{e^{inx}}{\sqrt{2\pi}}$ is orthonormal.

Definition 8.3

sps ϕ_n orthonormal. for riemann int'bl f, defn:

$$P_N f \ := \ \sum_{j=1}^{N} \phi_j \langle \phi_j, f\rangle$$

clearly $P_N(\alpha f + g) = \alpha P_N f + P_N g$, so P_N is in fact a lin op.

we show it is the ortho proj onto the span of $\{\phi_1, \ldots, \phi_N\}$.

Lemma 8.4

i) $P_N^2 f = P_N f$

ii) $\forall f, g$, we have:

$$\langle P_N f, g\rangle \ = \ \langle f, P_N g\rangle \ = \ \sum_{j=1}^{N} \langle f, \phi_j\rangle \langle \phi_j, g\rangle$$

in particular, P_N is self-adj.

Proof.

i) since ϕ_j orthonorm, note $P_N\phi_j = \phi_j$. then:

$$P_N P_N f \;=\; P_N \sum_{j=1}^{N} \phi_j \langle \phi_j, f\rangle \;=\; \sum_{j=1}^{N} \langle \phi_j, f\rangle P_N\phi_j \;=\; \sum_{j=1}^{N} \langle \phi_j, f\rangle \phi_j \;=\; P_N f$$

ii) by linearity:

$$\langle P_N f, g\rangle \;=\; \left\langle \sum_{j=1}^{N} \phi_j \langle \phi_j, f\rangle, g \right\rangle \;=\; \sum_{j=1}^{N} \langle f, \phi_j\rangle \langle \phi_j, g\rangle$$

it is easy to show this is equal to $\langle f, P_N g\rangle$.

■

Corollary 8.5

i) $\|P_N f\|^2 = \sum_{j=1}^{N} |\langle \phi_j, f\rangle|^2$

ii) $\langle P_N f, (1-P_N)g\rangle = 0$ for all f, g

Theorem 8.6

sps $f : [a,b] \longrightarrow \mathbb{C}$ riemann int'bl and ϕ_j orthonorm. let $c_k = \langle \phi_k, f\rangle$ and g_N a fn of the form:

$$g_N \;=\; \sum_{k=1}^{N} a_k \phi_k$$

for $a_k \in \mathbb{C}$. then:

i) $\|f - P_N f\| \leq \|f - g_N\|$ w equality iff $a_k = c_k$

ii) we have:

$$\|f\|^2 - \|f - g_N\| \;\leq\; \sum_{j=1}^{N} |c_j|^2 \;\leq\; \|f\|^2$$

Proof.

note $P_N g_N = g_N$.

i) note:

$$\begin{aligned}
\|f - g_N\|^2 &= \langle f - g_N, f - g_N \rangle \\
&= \langle f - P_N f + P_N f - g_N, f - P_N f + P_N f - g_N \rangle \\
&= \langle f - P_N f, f - P_N f \rangle + \langle f - P_N f, P_N f - g_N \rangle \\
&+ \langle P_N f - g_N f, f - P_N f \rangle + \langle P_N f - g_N, P_N f - g_N \rangle
\end{aligned}$$

by cor 7.5 above, this is equal to:

$$\begin{aligned}
\|f - g_N\|^2 &= \langle f - P_N f, f - P_N f \rangle + \langle P_N f - g_N, P_N f - g_N \rangle \\
&= \|f - P_N f\|^2 + \sum_{j=1}^{N} |a_j - c_j|^2 \qquad (*)
\end{aligned}$$

ii) notice:

$$\begin{aligned}
\|f - P_N f\|^2 &= \langle f - P_N f, f - P_N f \rangle \\
&= \langle f, f \rangle - \langle P_N f, f \rangle - \langle f, P_N f \rangle + \langle P_N f, P_N f \rangle \\
&= \langle f, f \rangle - \langle P_N f, P_N f \rangle \\
&= \|f\|^2 - \|P_N f\|^2
\end{aligned}$$

since $P_N^2 = P_N$ and it is self-adj. thus:

$$\|f - P_N f\|^2 \;=\; \|f\|^2 - \sum_{j=1}^{N} |c_j|^2 \;\geq\; 0 \qquad (**)$$

this gives the 2nd ineq. combining (*) and (**), we get:

$$\|f - g_N\|^2 \;=\; \|f\|^2 - \sum_{j=1}^{N} |c_j|^2 + \sum_{j=1}^{N} |a_j - c_j|^2 \;\leq\; \|f\|^2 - \sum_{j=1}^{N} |c_j|^2$$

giving us the first ineq.

■

Lec 14 - Mar 3 (Week 8)

wrote this parts notes early thinking i missed it. guess not lol

we now examine applications to fourier series. in particular, consider the case that $\phi_n = \frac{e^{inx}}{\sqrt{2\pi}}$, and:

$$P_N f \;=\; \sum_{n=-N}^{N} \phi_n \langle \phi_n, f \rangle$$

note we have to track factors of 2π. if we defn:

$$\widehat{f}(n) \;:=\; \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} \, \mathrm{d}x$$

then the fourier series assoc. to f is written as:

$$\sum_n \widehat{f}(n) e^{inx} \;=\; \sum_n \phi_n \langle \phi_n, f \rangle$$

so $\widehat{f}(n) = \frac{\langle \phi_n, f \rangle}{\sqrt{2\pi}}$.

Theorem 8.7: Parseval/Plancherel

sps $f : [-\pi, \pi] \longrightarrow \mathbb{C}$ riemann int'bl. then:

i) $\lim_{N\to\infty} \|f - P_N f\| = 0$

ii) we have:

$$\|f\|^2 \;=\; \sum_{\mathbb{Z}} |\langle \phi_n, f\rangle|^2 \;=\; 2\pi \sum_n \left|\widehat{f}(n)\right|^2 \;=\; \left(\int_{-\pi}^{\pi} |f(x)|^2 \,\mathrm{d}x\right)$$

[he wrote last one in class; not sure if acc the same.] we'll need the following lemma:

Lemma 8.8

he changed it...?

i) if f riemann int'bl, $\varepsilon > 0$, then $\exists g : (-\pi, \pi) \longrightarrow \mathbb{C}$ cts with $g(-\pi) = g(\pi)$ and $\|f - g\|_2 \leq \varepsilon$.

ii) if f riemann int'bl, $\varepsilon > 0$, then $\exists g_N = \sum_{k=-N}^{N} a_k \phi_k$ s.t. $\|f - g_N\|_2 < \varepsilon$.

note thm is immediate from ii).

Proof.

we start by proving ii) assuming i). let $\varepsilon > 0$. by i), $\exists g$ cts w $g(-\pi) = g(\pi)$ s.t. $\|f - g\|_2 < \varepsilon$. it suffices to approx g by $\sum_{k=-N}^{N} a_k \phi_k$.

by SW, if $h : (-\pi, \pi) \longrightarrow \mathbb{R}$ cts and $h(-\pi) = h(\pi)$, then $\exists$ trig poly p_N:

$$p_N \;=\; \sum_{n=1}^{N} a_n \cos(nx) + \sum_{n=1}^{N} b_n \sin(nx)$$

s.t. $\|p_N - h\|_\infty < \varepsilon$. but then:

$$\int_{-\pi}^{\pi} |p_N - h|^2 \,\mathrm{d}x \;\leq\; \varepsilon^2 2\pi$$

thus to approx g, we approx Re and Im parts separately, and note:

$$\sin(x) \;=\; \frac{e^{ix} - e^{-ix}}{2i} \qquad \cos(x) \;=\; \frac{e^{ix} + e^{-ix}}{2}$$

now, we provide sketch of i). by splitting into Re and Im parts, then +ve and -ve parts, assume $f \geq 0$ RI.

let $\varepsilon > 0$. since f riemann int'bl, $\exists$ partition $P = \{t_0, \ldots, t_n\}$ of $[-\pi, \pi]$ s.t. if:

$$M_j \;=\; \sup_{x \in [t_{j-1}, t_j]} f(x) \qquad m_j \;=\; \inf_{x \in [t_{j-1}, t_j]} f(x)$$

and:

$$f_L \;=\; \sum_{j=1}^{n} m_j \chi_{(t_{j-1}, t_j]} \qquad f_U \;=\; \sum_{j=1}^{n} M_j \chi_{(t_{j-1}, t_j]}$$

then:

$$\int_a^b |f_U - f| \, \mathrm{d}x \;\leq\; \int_a^b |f_U - f_L| \, \mathrm{d}x \;<\; \varepsilon$$

thus we can approx f_U by approxing χ over int'vls by trapezoids. ■

ok new stuff. uh. is it? recall:

$$\widehat{f}(n) \;=\; \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} \, \mathrm{d}x$$

then parseval says:

$$\sum_{|k| \leq N} \widehat{f}(k) e^{-ikx} \;\longrightarrow\; f$$

in mean square norm.

Theorem 8.9

sps f RI and cts at a pt $\theta_0 \in [-\pi, \pi]$. if $\widehat{f}(n) = 0 \ \forall n$, then $f(\theta_0) = 0$

indeed, if $\widehat{f}(n) = 0$, then $\int_{-\pi}^{\pi} |f(x)|^2 \, \mathrm{d}x = 0$.

Corollary 8.10

$$\sum_{\mathbb{Z}} \left| \widehat{f}(n) \right| < \infty \quad \Longrightarrow \quad \sum_{|k| \leq N} \widehat{f}(k) e^{-ikx} \rightrightarrows g$$

for some cts g. if f cts, then $f = g$ on $[-\pi, \pi]$.

Proof.

recall weierstrass M-text. if f_n fn's s.t. $\|f_n\|_\infty \leq M_n$ with $\sum M_n < \infty$, then:

$$\sum_{n=1}^{N} f_n \rightrightarrows F$$

for some F. the first part thus follows.

now, let $f_N = \sum_{|k| \leq N} \widehat{f}(k) e^{ikx}$. since $f_N \rightrightarrows g$, we know:

$$\begin{aligned} \widehat{g}(k) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} g(x) e^{-ikx} \, \mathrm{d}x = \frac{1}{2\pi} \int_{-\pi}^{\pi} \lim_{N \to \infty} f_N(x) e^{-ikx} \, \mathrm{d}x \\ &= \lim_{N \to \infty} \frac{1}{2\pi} \int_{-\pi}^{\pi} f_N(x) e^{-ikx} \, \mathrm{d}x \\ &= \widehat{f}(k) \end{aligned}$$

thus $\widehat{g}(k) = \widehat{f}(k)$ for all k, so $h = f - g$ cts and $\widehat{h}(k) = \widehat{f}(k) - \widehat{g}(k) = 0$, so $h = 0$. ■

Lec 15 - Mar 5 (Week 8)

sub tdy weeee

oh he switches the order of the inner prod terms:

$$\langle f, g\rangle \;=\; \int_{-\pi}^{\pi} f(x)\overline{g(x)}\;\mathrm{d}x$$

..? sps f diff'ble, f' int'bl, $f(-\pi) = f(\pi)$. then using IBP:

$$\begin{aligned}\widehat{f}(k) \;=\; \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-ikx}\;\mathrm{d}x \;&=\; \frac{-1}{2\pi}\int_{-\pi}^{\pi} f'(x)\frac{1}{-ik}e^{-ikx}\;\mathrm{d}x \\ &=\; \frac{1}{ik}\frac{1}{2\pi}\int_{-\pi}^{\pi} f'(x)e^{-ikx}\;\mathrm{d}x \\ &=\; \frac{1}{ik}\widehat{f'}(k)\end{aligned}$$

so:

$$\widehat{f}(k) = \frac{1}{ik}\widehat{f'}(k) \quad\Longrightarrow\quad k\left|\widehat{f}(k)\right| = \left|\widehat{f'}(k)\right|$$

now, sps f is m-times diff'ble, $f^{(k)}(-\pi) = f^{(k)}(\pi)$. doing IBP m times gives:

$$\left|\widehat{f}(k)\right| \;=\; \frac{1}{k^m}\left|\widehat{f}^{(m)}(k)\right| \;\leq\; \frac{C}{k^m}$$

where $\left|\widehat{f}^{(m)}\right| \leq C$. exercise(?): $\widehat{(f')} \sim k\widehat{f}$.

III Measure Theory

9 Basics

riemann integral =(. "im kind of biased in saying this, but im also correct, so". fire.

Definition 9.1

for an int'vl $I \subseteq \mathbb{R}$ w endpts $a < b$, define $\ell(I) = b - a$. for $A \subseteq \mathbb{R}$, its **outer measure** $m^*(A)$ is:

$$m^*(A) := \inf \left\{ \sum_{k=1}^{\infty} \ell(I_k) : A \subseteq \bigcup_{k=1}^{\infty} I_k,\ I_k \text{ open bdd int'vls} \right\}$$

note $m^*(A) \in [0, \infty]$, i.e. $\mathbb{R}_{\geq 0}$ but can also be infty.

Proposition 9.2

i) if $A \subseteq B$, $m^*(A) \leq m^*(B)$

ii) if E ctbl, $m^*(E) = 0$

iii) if I int'vl, then $m^*(I) = \ell(I)$

Proof.

i) any cvr of B cvrs A.

ii) if $E = \{x_i\}$, let $I_k = (x_k - \varepsilon/2^{k+1}, x_k + \varepsilon/2^{k+1})$. then:

$$m^*(E) \leq \sum \ell(I_k) = \sum \varepsilon/2^k = \varepsilon$$

iii) sps $J = [a, b]$ (renamed). sps I_k open int'vls cvring J. J cpt, so take finitely

many; also assume no $I_j \subseteq I_k$. if $I_k = (a_k, b_k)$ and $\{a_i\}, \{b_i\}$ increasing, then:

$$\sum_{j=1}^{k} \ell(I_j) \;=\; \sum_{j=1}^{k} b_j - a_j \;=\; b_k - a_1 + \sum b_j - a_{j+1} \;>\; b_k - a_1 \;\geq\; b - a$$

so $\forall$ cvrs $\{I_j\}$, $\sum \ell(I_j) \geq b - a$ implies $m^*(J) \geq b - a$. OTOH, if $J_\varepsilon = (a - \varepsilon, b + \varepsilon)$, then $m^*(J) \geq \ell(J_\varepsilon) \to b - a$. [i love aboos of notation]

now, if $J = (a, b)$, let $I^- = [a + \varepsilon, b - \varepsilon]$ and $I^+ = [a - \varepsilon, b + \varepsilon]$. clearly, $I^- \subseteq J \subseteq I^+$, so $m^*(I^-) \leq m^*(J) \leq m^*(I^+)$, and it follows.

■

Lemma 9.3

if $E \subseteq \mathbb{R}$ and $y \in \mathbb{R}$, then $m^*(E + y) = m^*(E)$.

translate ur cvrs.

Lemma 9.4

if $E \subseteq \bigcup_{k=1}^{\infty} E_k$, then $m^*(E) \leq \sum m^*(E_k)$.

Proof.

for all k, let $\left\{I_j^{(k)}\right\}$ be an int'vl cvr of E_k s.t.:

$$m^*(E_k) \;\leq\; \sum_{j=1}^{\infty} \ell(I_j^{(k)}) \;\leq\; m^*(E_k) + \frac{\varepsilon}{2k}$$

since $\left\{I_j^{(k)}\right\}_{j,k}$ an int'vl cvr of E, then:

$$m^*(E) \;\leq\; \sum_{k=1}^{\infty}\sum_{j=1}^{\infty}\ell(I_j^{(k)}) \;\leq\; \sum_k m^*(E_k) + \frac{\varepsilon}{2k} \;\leq\; \sum_k m^*(E_k) \;+\; \varepsilon$$

■

Definition 9.5

a set $E \subseteq \mathbb{R}$ is **measurable** if $\forall A \subseteq \mathbb{R}$:

$$m^*(A) \;=\; m^*(A \cap E) + m^*(A \cap E^c)$$

we denote by $\mathcal{E}$ the set of measurable sets.

note by the union property, we always have $m^*(A) \leq m^*(A \cap E) + m^*(A \cap E^c)$. thus, to show E measurable, we only need to show:

$$m^*(A) \;\geq\; m^*(A \cap E) + m^*(A \cap E^c)$$

note also, non-measurable sets exist; that is, $\exists A, B$ disjoint s.t.:

$$m^*(A \cup B) \;\lneq\; m^*(A) + m^*(B)$$

Lemma 9.6

if $m^*(E) = 0$, then $E \in \mathcal{E}$.

Proof.

note $m^*(A \cap E) + m^*(A \cap E^c) \;\leq\; m^*(E) + m^*(A) \;=\; m^*(A)$. ■

Lemma 9.7

if $E_1, E_2 \in \mathcal{E}$, then $E_1 \cup E_2 \in \mathcal{E}$.

Proof.

sps $A \subseteq \mathbb{R}$. then:

$$\begin{aligned} m^*(A) &= m^*(A \cap E_1) + m^*(A \cap E_1^c) \\ &= m^*(A \cap E_1) + m^*(A \cap E_1^c \cap E_2) + m^*(A \cap E_1^c \cap E_2^c) \\ &\geq m^*(A \cap (E_1 \cup E_2)) + m^*(A \cap (E_1 \cup E_2)^c) \end{aligned}$$

since $A \cap (E_1 \cup E_2) \;=\; (A \cap E_1) \cup (A \cap E_1^c \cap E_2)$. ■

Definition 9.8

a coll'n $\mathcal{A} \subseteq \mathcal{P}(\mathbb{R})$ is an **algebra** if:

- $\mathbb{R} \in \mathcal{A}$
- $E \in \mathcal{A} \implies E^c \in \mathcal{A}$
- $E_1 \cup E_2 \in \mathcal{A} \implies E_1 \cup E_2 \in \mathcal{A}$

Corollary 9.9

$\mathcal{E}$ is an alg

Proposition 9.10

if $C_1, C_2 \in \mathcal{E}$ disj, then $m^*(C_1 \cup C_2) = m^*(C_1) + m^*(C_2)$.

Proof.

note $A \cap E_1 \;=\; A \cap E_1 \cap E_2 + A \cap E_1 \cap E_2^c$, so:

$$\begin{aligned} m^*(A \cap (E_1 \cup E_2)) &= m^*(A \cap (E_1 \cup E_2) \cap E_2) + m^*(A \cap (E_1 \cup E_2) \cap E_2^c) \\ &= m^*(A \cap E_1) + m^*(A \cap E_2) \end{aligned}$$

■

Proposition 9.11

$\mathcal{E}$ closed under ctbl unions.

Proof.

let $A_j \in \mathcal{E}$. denote:

$$E_1 \;=\; A_1 \qquad E_j \;=\; A_j \setminus \bigcup_{k=1}^{j-1} A_k$$

then E_j disj and $E = \bigcup E_j$. let $F_n = \bigcup_{j=1}^n E_j$. then:

$$\begin{aligned} m^*(A) &= m^*(A \cap F_n) + m^*(A \cap F_n^c) \\ &\geq m^*(A \cap F_n) + m^*(A \cap E^c) \\ &= \sum_{j=1}^n m^*(A \cap E_j) + m^*(A \cap E^c) \end{aligned}$$

taking $n \to \infty$:

$$m^*(A) \;\geq\; \sum_{j=1}^\infty m^*(A \cap E_j) + m^*(A \cap E^c) \;\geq\; m^*(A \cap E) + m^*(A \cap E^c)$$

■

Definition 9.12

a σ-algebra is an alg closed under ctbl unions.

thus, $\mathcal{E}$ is a σ-alg. note also that $\mathcal{E}$ closed under ctbl intersections by taking complements.

Definition 9.13

we call $\mathcal{E}$ the set of **Lebesgue measurable** sets, and denote $m := m^*|_{\mathcal{E}}$ as the **Lebesgue measure**.

Lemma 9.14

$(a, \infty) \in \mathcal{E}$.

Proof.

let $A \subseteq \mathbb{R}$; first sps $a \notin A$.

sps $\{I_j\}$ cvrs A. let $J_k = I_k \cap (-\infty, a), K_k = I_k \cap (a, \infty)$. note:

$$\sum_j \ell(I_j) \;=\; \sum_j \ell(J_j) + \sum_j \ell(K_j)$$

we also have:

$$A \cap (-\infty, a) \;\subseteq\; \bigcup_j J_j \qquad A \cap (a, \infty) \;\subseteq\; \bigcup_j K_j$$

therefore, we have that:

$$m^*(A \cap (-\infty, a)) + m^*(A \cap (a, \infty)) \;\leq\; \sum_j \ell(J_k) + \sum_j \ell(K_j) \;=\; \sum_j \ell(I_j)$$

and it follows that $m^*(A \cap (-\infty, a)) + m^*(A \cap (a, \infty)) \leq m^*(A)$. ■

Lec 16 - Mar 10 (Week 9)

(missed)

Corollary 9.15

all open, half-open, closed int'vls are in $\mathcal{E}$

Proof.

note:

$$[a, \infty) \;=\; \bigcap_{n \geq 1} (a - n^{-1}, \infty) \;\in\; \mathcal{E}$$

thus, any semi-infinite int'vl in $\mathcal{E}$, so $(a, b) = (a, \infty) \cap (-\infty, b)$ in $\mathcal{E}$. other int'vls follow similarly. ■

Corollary 9.16

any open set is in $\mathcal{E}$

Theorem 9.17: Regularity

let $E \in \mathcal{E}$, $\varepsilon > 0$. then there are open and closed O, F resp., s.t.:

$$m(O \setminus E) < \varepsilon \qquad m(E \setminus F) < \varepsilon$$

Proof.

we start with outer reg. assume $m(E) < \infty$. then there are int'vls I_k s.t.:

$$\sum_k m(I_k) \ < \ m(E) + \varepsilon$$

then $O = \bigcup_k I_k$ works since:

$$m(O \setminus E) \ = \ m(O) - m(E) \ \leq \ \sum_k m(I_k) - m(E) \ < \ \varepsilon$$

now sps $m(E) = \infty$. let $E_k = E \cap [-k, k]$. then $m(E_k) \leq 2k$. choose sets $O_k \supseteq E_k$ s.t. $m(O_k \setminus E_k) < \frac{\varepsilon}{2^k}$. let $O = \bigcup_k O_k$. then:

$$m(O \setminus E) \ \leq \ \sum_k m(O_k \setminus E_k) \ < \ \varepsilon$$

for inner reg., choose $O \supseteq E^c$ s.t. $m(O \setminus E^c) < \varepsilon$ and let $F = O^c$. then:

$$m(E \setminus F) \ = \ m(O \setminus E^c) \ < \ \varepsilon$$

since $E \setminus F \ = \ E \cap F^c \ = \ E \cap O \ = \ O \setminus E^c$. ■

Corollary 9.18

for any $E \in \mathcal{E}$, there are open $O_n \supseteq E$ and closed $F_n \subseteq E$ s.t.:

$$m\left(\bigcap_n O_n \setminus E\right) = 0 \qquad m\left(E \setminus \bigcup_n F_n\right) = 0$$

this is another way of saying that we measure a set by an open outer approx or closed inner approx. here, we want the measure of the difference to go to 0.

Lec 17 - Mar 17 (Week 10)

midterm last thurs

Theorem 9.19

i) if $A_n \subseteq A_{n+1} \in \mathcal{E}$ then $m\left(\bigcup A_n\right) = \lim_n m(A_n)$

ii) if $m(B_1) < \infty$ and $B_n \supseteq B_{n+1}$ then $m\left(\bigcap B_n\right) = \lim_n m(B_n)$

Proof.

i) let $C_1 = A_1$ and $C_n = A_n \setminus A_{n-1} = A_n \cap (A_{n-1})^c \in \mathcal{E}$. then they are disj, so:

$$m\left(\bigcup_{n=1}^{\infty} A_n\right) = m\left(\bigcup_{n=1}^{\infty} C_n\right) = \sum_{n=1}^{\infty} m(C_n)$$

$$= \lim_{n\to\infty} \sum_{k=1}^{n} m(C_k) = \lim_{n\to\infty} m\left(\bigcup_{k=1}^{n} C_k\right) = \lim_{n\to\infty} m(A_n)$$

ii) let $A_n = B_1 \setminus B_n$. since B_n is decreasing, A_n is increasing, so

$$m\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} m(A_n)$$

note:

$$m(A_n) = m(B_1 \setminus B_n) = m(B_1) - m(B_n)$$

OTOH:

$$\begin{aligned} m\left(\bigcup_{n=1}^{\infty} A_n\right) &= m\left(\bigcup_{n=1}^{\infty} B_1 \setminus B_n\right) \\ &= m\left(B_1 \setminus \bigcap_{n=1}^{\infty} B_n\right) \\ &= m(B_1) - m\left(\bigcap_{n=1}^{\infty} B_n\right) \\ &= \lim_{n\to\infty} m(B_1) - m(B_n) \end{aligned}$$

■

recall the defn of a σ-alg. given a coll'n of subsets of $\mathbb{R}$ the σ-alg *generated by* A is the intersection of all σ-algs containing A, denoted $\sigma(A)$.

Definition 9.20

a **measure space** $(\Omega, \mathcal{F})$ is a set Ω and a σ-alg $\mathcal{F}$ on Ω. given two measure spaces $(\Omega_1, \mathcal{F}_1), (\Omega_2, \mathcal{F}_2)$, we say $f : \Omega_1 \longrightarrow \Omega_2$ is **measurable** if $\forall E \in \mathcal{F}_2$, we have that $f^{-1}(E) \in \mathcal{F}_1$.

note the similarity to (topological) continuity, not that the similarity is particularly important though.

Lemma 9.21

if $(\Omega_1, \mathcal{F}_1), (\Omega_2, \mathcal{F}_2)$ are measure spaces s.t. $\mathcal{F}_2 = \sigma(A)$ for some coll'n $A \subseteq \mathcal{P}(\Omega_2)$, then $f : \Omega_1 \longrightarrow \Omega_2$ is measurable if:

$$f^{-1}(E) \in \mathcal{F}_1$$

for all $E \in A$.

Proof.

suffices to check that

$$S := \{E : f^{-1}(E) \in \mathcal{F}_1\}$$

is a σ-alg containing A. indeed, its a σ-alg essentially bc $\mathcal{F}_1$ is.

$$E_n \in S \implies f^{-1}\left(\bigcup_{n=1}^{\infty} E_n\right) = \bigcup_{n=1}^{\infty} f^{-1}(E_n) \in \mathcal{F}_1 \implies \bigcup_{n=1}^{\infty} E_n \in S$$

other axioms are similar. ■

so why do we care abt this lemma?

Definition 9.22

the **Borel $\boldsymbol{\sigma}$-alg** is given as $\mathcal{B}(\mathbb{R}) := \sigma(\text{open subsets of } \mathbb{R})$.

fact: $\mathcal{B}(\mathbb{R}) \subsetneq \mathcal{E} \subsetneq 2^{\mathbb{R}}$. the inclusions being strict is tedious and unnecessary to show, and thus skipped.

Definition 9.23

we say a fn $f : \mathbb{R} \longrightarrow \mathbb{R}$ is **Lebesgue measurable** if it is measurable as a fn $(\mathbb{R}, \mathcal{E}) \longrightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$. we say it is **Borel measurable** if it is measurable as a fn $(\mathbb{R}, \mathcal{B}(\mathbb{R})) \longrightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Lemma 9.24

a fn $f : \mathbb{R} \longrightarrow \mathbb{R}$ is lebesgue (borel) meas if:

$$f^{-1}((c, \infty)) \in \mathcal{E} \qquad (\text{or } \mathcal{B}(\mathbb{R}))$$

for any $c \in \mathbb{Q}$.

Proof.

suffices to check $\{\text{open sets}\} \subseteq \sigma(\{(c, \infty)\}) =: \mathcal{F}$ for $c \in \mathbb{Q}$.

i) check $(a, b) \in \mathcal{F}$. indeed, $(a, b) = (a, \infty) \setminus [b, \infty)$, but $[b, \infty) = \bigcap (q, \infty)$ for $q \in \mathbb{Q} \cap (-\infty, b)$.

ii) any open set in $\mathbb{R}$ is a ctbl union of open intvls w/ rational endpts.

■

Corollary 9.25

any cts or monotonic $f : \mathbb{R} \longrightarrow \mathbb{R}$ is borel measurable.

Proposition 9.26

sps $f, g : \mathbb{R} \longrightarrow \mathbb{R}$ are lebesgue (borel) meas. the following are as well:

i) $|f|, \alpha f + \beta$ for real α, β

ii) $(f)_+, (f)_-$ where $(x)_*$ is the $+$ve or $-$ve parts of x

Proof.

these are compositions of f w/ a cts fn, hence borel meas. easy to see that the composition of a BM fn w/ a LM fn is LM.

■

Lec 18 - Mar 19 (Week 10)

Proposition 9.27

in addition to the above, the following are also lebesgue (borel) meas.:

iii) $f + g, f - g, fg$

iv) $\max(f, g), \min(f, g)$

Proof.

iii) we show $f+g$ is lebesgue. suffices to check $\{x : f(x)+g(x) > c\} \in \mathcal{E}\ \forall c \in \mathbb{R}$. we claim that:

$$\{x : f(x)+g(x) > c\} \in \mathcal{E} \ = \ \bigcup_{q \in \mathbb{Q}} \{x : f(x) > c-q\} \cap \{x : g(x) > q\}$$

enough to check $(\subseteq)$. indeed, sps x is s.t. $f(x)+g(x) > c$. then $\exists q \in \mathbb{Q}$ with $g(x) > q > c - f(x)$, with $f(x) > c-q$ and $g(x) > q$.

$f-g$ is similar, so we show fg. note:

$$fg \ = \ \frac{1}{4}\left((f+g)^2 - (f-g)^2\right)$$

wts if f meas, then f^2 is, but this is just a composition of meas. fns.

iv) follows from above, as:

$$\max(f,g) \ = \ \frac{|f-g|+(f+g)}{2} \qquad \min(f,g) \ = \ \frac{(f+g)-|f-g|}{2}$$

■

it is sometimes useful to consider fns which take values $\pm\infty$.

Definition 9.28

let $\overline{\mathbb{R}} = \mathbb{R} \cup \{\pm\infty\}$ denote the extended real line. we define:

$$\mathcal{B}(\overline{\mathbb{R}}) \ := \ \left\{E \subseteq \overline{\mathbb{R}} : E = A \cup B, A \in \mathcal{B}(\mathbb{R}), B \in \{\varnothing, \{\infty\}, \{-\infty\}, \{\pm\infty\}\}\right\}$$

Exercise 9.29

i) $\mathcal{B}(\overline{\mathbb{R}})$ is a σ-alg

ii) $f : \mathbb{R} \longrightarrow \overline{\mathbb{R}}$ is LM (BM) iff $\{x : f(x) > c\} \in \mathcal{E}$ or $\mathcal{B}(\mathbb{R})$ $\forall c \in \mathbb{Q}$

iii) if $f : \mathbb{R} \longrightarrow \mathbb{R}$ meas., then $\iota \circ f : \mathbb{R} \longrightarrow \overline{\mathbb{R}}$ meas.

any meas. fn defined on the ext. reals is (unsurprisingly) also called LM/BM. the difference only rly matters when it comes to operations on fns (whats $\infty - \infty$?).

given $E \in \mathcal{E}$, $f : E \longrightarrow \overline{\mathbb{R}}$, tfae:

- f meas
- $f^{-1}(A) \in \mathcal{E}$ or $\mathcal{B}(\mathbb{R})$ $\forall A \in \mathcal{B}(\overline{\mathbb{R}})$
- the fn $\widetilde{f} = \chi_E f$ is meas.

equivalence of last one is an exercise.

10 Limits of measurable functions

Proposition 10.1

let $f_n : E \longrightarrow \overline{\mathbb{R}}$ a seq of ptwise incresing meas. fns for some E meas. then $\lim_{n\to\infty} f_n(x) = f(x)$ is meas.

Proof.

for all $c \in \mathbb{R}$, since f_n is ptwise increasing, we have:

$$\{x : f(x) > c\} = \bigcap_{n=1}^{\infty} \{x : f_n(x) > c\}$$

■

Theorem 10.2

sps E meas., $f_n : E \longrightarrow \overline{\mathbb{R}}$ seq of meas. fns. the following are also meas.:

i) $x \mapsto \sup_n f_n(x)$

ii) $x \mapsto \inf_n f_n(x)$

iii) $x \mapsto \limsup_n f_n(x)$

iv) $x \mapsto \liminf_n f_n(x)$

v) $x \mapsto \lim_n f_n(x)$, provided this exists $\forall x$

Proof.

i) $\forall c \in \mathbb{R}$:

$$\left\{x : \sup_n f_n(x) > c\right\} = \bigcup_{n=1}^{\infty} \{x : f_n(x) > c\}$$

ii) similar as above

iii) recall $\limsup_n f_n(x) = \lim_n \left\{\sup_{m \geq n} f_m(x)\right\}$. this is a decreasing seq of meas. fns, so by i), we're done

iv) similar as above

v) if the lim exists, its equal to both limsup and liminf

■

we know what a.e. is (uses outer meas. specifically, since we dk a priori the set is meas)

Lemma 10.3

sps $f : \mathbb{R} \longrightarrow \overline{\mathbb{R}}$ LM and $g : \mathbb{R} \longrightarrow \overline{\mathbb{R}}$ is s.t. $f(x) = g(x)$ ae. then g meas.

Proof.

let $E = \{x : f(x) \neq g(x)\}$. for $A \in \mathcal{B}(\overline{\mathbb{R}})$, we have:

$$g^{-1}(A) \;=\; (f^{-1}(A) \cap E^c) \cup F$$

where $F = g^{-1}(A) \cap E$. then $m^*(F) \leq m^*(E) = 0$, so $F \in \mathcal{E}$. by assumption, $f^{-1}(A) \in \mathcal{E}$ and $E^c \in \mathcal{E}$, so $g^{-1}(A) \in \mathcal{E}$. ■

Corollary 10.4

sps $f_n : \mathbb{R} \longrightarrow \overline{\mathbb{R}}$ meas s.t. $\lim_n f_n(x)$ exists ae. let $f(x) = \lim_n f_n(x)$ when it exists, and arbitrarily otw. then f LM.

Proof.

let $E = \{x : \lim_n f_n(x) \text{ exists}\}$. note $E \in \mathcal{E}$ since $m^*(E^c) = 0$. defn:

$$\widetilde{f_n}(x) \;=\; \begin{cases} f_n(x) & x \in E \\ 0 & \text{otw} \end{cases}$$

then $\widetilde{f}(x) = \lim_n \widetilde{f_n}(x)$ exists $\forall \mathbb{R}$. note:

i) $\widetilde{f_n}(x)$ LM since $\widetilde{f_n} = \chi_E f_n$

ii) $\widetilde{f}$ LM since its ptwise lim of LM fns

thus $f = \widetilde{f}$ a.e., so by lemma, f is LM. ■

Exercise 10.5

show E defn'd as above is always meas., even if the lim doesnt necessarily exist ae.

hint: lim exists iff limsup = liminf.

11 Simple functions

throughout this section, $E \in \mathcal{E}$ unless stated otw.

Definition 11.1

an LM $\phi : E \longrightarrow \mathbb{R}$ is **simple** if $|\mathrm{im}(\phi)| < \infty$. equivalently:

$$\phi = \sum_{k=1}^{n} \alpha_k \chi_{E_k}$$

where $\alpha_k \in \mathbb{R}, E_k \in \mathcal{E}$.

Lemma 11.2

sps $f : E \longrightarrow \mathbb{R}$ bdd LM, $\varepsilon > 0$. then $\exists \phi_\varepsilon, \psi_e$ simple s.t. $\phi_\varepsilon \leq f \leq \psi_\varepsilon$, and $|\psi_e - \phi_e| < \varepsilon$ for all x.

Proof.

let $M > |f(x)|$ for all $x \in E$, and write:

$$(-M, M] = \bigcup_{j=1}^{n} (y_{j-1}, y_j]$$

s.t. $|y_j - y_{j-1}| < \varepsilon$ for some strictly increasing y_j. let $E_j = f^{-1}((y_{j-1}, y_j])$. by choice of M:

$$E = \bigsqcup_{j=1}^{n} E_j \qquad \phi_\varepsilon = \sum_{j=1}^{n} y_{j-1} \chi_{E_j} \qquad \psi_\varepsilon = \sum_{j=1}^{n} y_j \chi_{E_j}$$

■

Lemma 11.3

sps $f : E \longrightarrow \overline{\mathbb{R}}$ LM. then, $\exists$ seq of simple $\phi_n \to f$ ptwise s.t. $|\phi_n| \leq |f|$.

if $f \geq 0$, then furthermore ϕ_n non-neg increasing.

Proof.

general follows from non-neg case: note that we have $\phi_n \to (f)_+, \psi_n \to (f)_-$ with $\phi_n \leq (f)_+, \psi_n \leq (f)_-$. thus $\phi_n - \psi_n \to (f)_+ - (f)_- = f$, and:

$$|\phi_n - \psi_n| = \phi_n + \psi_n \leq (f)_+ + (f)_- = |f|$$

for $f \geq 0$, consider $f_n(x) = \min(f(x), n)$ LM. by lemma, $\exists \phi_n$ simple s.t. $\phi_n \leq f_n$ and $|f_n(x) - \phi_n(x)| \leq 1/n$. let $\psi_1 = \max(\phi_1, 0)$, and $\psi_n = \max(\phi_n, \psi_{n-1})$. claim:

i) ψ_n are simple: by ind, ψ_n is max of two simples thus simple (fairly clear)

ii) ψ_n are increasing: clear

iii) $\psi_n \to f$: by construction + ind, $\phi_n \leq \psi_n \leq f_n \leq f$ for all n

■